

Budgets as Grades: Which Equal-Meaning Inputs a Budgeted Reasoner Distinguishes

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Abstract

A recognizer that answers Horn entailment queries with a fixed computation budget does not treat logically identical inputs alike. We ask *which* of them it separates, and answer with a graded object. The closure operator of propositional Horn logic factors into stages T_k , the atoms derivable in k parallel rounds of forward chaining, with $T_0 = \text{id}$, $T_j \circ T_k = T_{j+k}$, and limit the closure. We prove in Lean 4, without Mathlib, that the limit is sound and complete, that budget- k behavioural identity is blind to premise order and repetition at every grade, and that a trace is separated from a redundant extension at budget k exactly when the extension moves some derivation across the budget. This yields a preregistered prediction: a derivable clause that shortens the target derivation changes a budgeted recognizer’s decision at a low budget and not at a high one, while an equally redundant clause that preserves depth changes nothing at any budget. On 200 fresh cases the prediction held for a learned iterative reasoner (interaction 0.885, 95% CI [0.84, 0.93]) whose decisions coincided with the k -round symbolic reasoner at every budget. A second, independent prediction, the composition law read on inputs (“a hint buys exactly its depth”), was then tested on the same weights and failed at tight budgets (agreement 0.90 and 0.83 against a preregistered 0.95 criterion), while holding for a model trained under a smaller budget; every violation was a lost derivation. A third preregistered test on a further 200 fresh cases confirmed the resulting one-round sandwich law without exception: the learned operator’s answer from j -round-saturated hypotheses at budget k lies between the symbolic answers at budgets $j+k-1$ and $j+k$, so it is a lax graded reader with exactly one round of slack. Two open-weight language models (0.5B and 1.5B parameters) did not read the table at the decision level; at the margin level their sensitivity to redundant clauses was dominated by lexical overlap with the query. A language model with a thinking budget (Gemini 3.1 Flash-Lite) is the exact oracle on the table once it thinks; without thinking, the preregistered unsigned test failed, and a preregistered follow-up on a further fresh set of 200 cases showed why: a depth-shortening redundant clause raises its correct-answer rate by 0.155 and an equally redundant depth-preserving clause lowers it by 0.070, two effects of opposite sign that cancel in an unsigned statistic and both vanish once the model can spend tokens. A fourth preregistered test found the cost to grow with the number of redundant clauses, an irrelevant clause to cost three times a redundant one, and the budget response to be a step rather than a slope. Fed its own outputs, the learned reader composes within the bounds the lax structure proves (slack at most two rounds, usually one), is monotone, and has no unit. The graded structure therefore describes what a budgeted reasoner distinguishes; what training keeps of the graded monad is a lax, unitless graded family of monotone operators inside it; and for a language model at zero budget the distinguishing variable is two-sided, depth and clause count, and was found by testing laws rather than by assuming them.

Introduction

Two inputs with the same meaning are, to a recognizer, two inputs. The recognition-paths framework [1] makes this precise for propositional Horn logic. A *recognizer* is any map from an ordered premise trace and a query to an observation. It induces a behavioural identity $\approx_\rho$ on traces (no context and no query separates them); logical identity $\equiv_L$ is theory equivalence (the same consequences for every query); both are congruences on the free monoid of traces, and the recognizer *respects logic* exactly when $\equiv_L$ refines $\approx_\rho$, equivalently when a unique monoid morphism from the logical quotient to the behavioural quotient exists (the Recognition Factorization Theorem, machine-checked in Lean).

The empirical half of that programme [2] found that no recognizer examined respects logic: language models up to 1.5B parameters and small constructed models alike distinguish premise permutations, repetitions, and, most robustly, a trace from its extension by a *derivable* clause, a clause that changes no consequence. A set encoder that is permutation-invariant by architecture is exactly the syntactic quotient and only partly the semantic one; an equivalence objective does not change this. That is a negative result about identification.

This paper asks the positive question it leaves open. If a recognizer does not identify all logically identical inputs, *which* of them does it separate, and is there a mathematical object that predicts the answer rather than describes it after the fact? The programme’s stated rule is that the object must come after an observed law, not before: observation, then equivalence, then quotient, then operations and equations, then an algebraic theory, and only then a monad. The results of [2] gave the first three. This paper supplies an object that generates equations, tests two of them under preregistration, and reports that one held and one failed.

The object is the *graded closure*. Forward chaining, stopped after k parallel rounds, is an operator T_k on sets of atoms. The family $(T_k)_{k \in \mathbb{N}}$ has $T_0 = \text{id}$ and $T_j \circ T_k = T_{j+k}$; it is an $\mathbb{N}$ -graded monad on the poset of atom sets [3,4] whose limit T_∞ is the closure operator of Horn logic. Only the limit is idempotent. Reading the grade as a *computation budget* turns the algebra into predictions about recognizers with a budget knob.

Contributions.

1. *Theory*. A Lean 4 development (core library only) of the graded closure: the graded-monad laws, soundness and completeness of the limit, a budget- k behavioural identity $\approx_k$ that is blind to order and repetition at every grade, the theorem that a trace and its extension by one clause are separated at budget k if and only if the extension moves some query’s derivation from depth $> k$ to depth $\leq k$, and the composition law read on inputs: pre-saturating the hypotheses by j rounds and reading with budget k equals reading with budget $j + k$.
2. *A held prediction*. The separation theorem, preregistered as an interaction between two budgets on a frozen table of 200 fresh Horn cases, held for a learned iterative reasoner with a rounds budget, with the full predicted shape and with exact agreement with the symbolic k -round reasoner at every budget.
3. *A failed prediction, and the law that replaces it*. The composition law, preregistered on the same weights, failed at the two tight budget pairs and held at the two others; all violations were lost derivations from multi-atom hypothesis sets. A model trained under a smaller budget satisfied the law. The one-round sandwich law suggested by the failure was then preregistered and confirmed on a further 200 fresh cases on every pair and every case: the learned reasoner is exactly graded in what it separates and laxly graded, with one round of slack, in how it composes.
4. *The monad question, answered for the budgeted reasoner*. Logical identity is equality of closures under the closure monad on theories (Lean, Closure.lean); the graded closure is its filtration. A *lax graded reader* (monotone operators sandwiched by the graded closure with slack s) is defined and its composition laws proved. Fed its own outputs on 200 fresh cases, the learned reader satisfies the lax composition law (slack ≤ 2 , preregistered; slack 1 on eight of nine pairs), is monotone (preregistered), and fails the unit law. The algebraic object the data support for a trained budgeted reasoner is a lax, unitless graded family inside the graded monad, not the monad.
5. *Open-weight language models*. At 0.5B and 1.5B parameters, single forward pass, the models do not read the table at the decision level; on the margin level the variable that predicts their sensitivity to redundant clauses is lexical overlap with the query, not depth.
6. *A language model with a thinking budget*. With a thinking budget of 1024 tokens, Gemini 3.1 Flash-Lite has the oracle’s decision profile on every row of two 200-case tables. Without thinking, the preregistered unsigned interaction failed; the preregistered signed follow-up on a fresh table held: a shortening clause helps (+0.155), a depth-preserving redundant clause hurts (−0.070), and thinking removes both effects exactly. A dose–response test then showed the cost growing with the number of

redundant clauses, an irrelevant clause costing three times a redundant one (the preregistered “count” expectation was wrong), and the budget acting as a threshold: about 140 thinking tokens change nothing and about 300 make the model the oracle.

Everything is frozen (SHA-256 locks on tables, commit hashes on code, weights on Hugging Face with training logs) and every learned-model result was preceded by a committed preregistration naming the comparison, the criterion, the controls and the sentence to be reported on failure.

Related work

Order sensitivity of language models. Chen et al. [5] showed that premise order changes accuracy on deductive tasks even when the logic is unchanged; metamorphic and reformulation testing [6,7] and parameterized logical benchmarks [8] measure the same phenomenon through accuracy differences. Our instrument reads order-*relations* on the answer logits so that identity is exact rather than approximate, and compares language models and constructed recognizers on the same frozen table [2]. Set-LLM [9] and order-centric augmentation [10] aim at permutation invariance; the companion paper showed that permutation invariance is strictly weaker than logical invariance, and the present one shows that the gap has a graded structure.

Depth and computation. PrOntoQA [11] and related synthetic suites vary proof depth and find accuracy falling with depth; Merrill and Sabharwal [12] bound what transformers compute per step and what chain of thought adds; Barceló et al. [13] characterise message-passing networks by the logic of k -hop neighbourhoods. Our reasoner is a message-passing network by design, so the one-hop-per-round bound is part of its specification, and we say so in advance. What is tested is whether training realises the graded reader rather than a shortcut, whether the gap closes exactly at the predicted budget, and whether the *composition* law, which no locality bound implies, holds.

Graded monads and closure operators. Graded (parametric effect) monads [3,4,14] index a monad by a monoid; the closure operator of Horn logic is the classical least-model construction [15], and its truncation at k rounds is the standard bounded forward chaining. The novelty is not the algebra, which is elementary, but its use as a source of *falsifiable equations* about learned recognizers, and the observation that one equation is a property of architecture (identification) while the other is a property of training (composition).

Automata and identification. Behavioural identity as Nerode equivalence and finite-test identification as closure [16,17] underlie the companion framework; graded identity $\approx_k$ is a family of such equivalences indexed by budget, coarser than the syntactic quotient and finer than $\equiv_L$ for every finite k .

Setting

We recall the definitions of [1] that are used here. Fix a type of atoms. A Horn clause is $a_1 \wedge \dots \wedge a_m \rightarrow b_1 \wedge \dots \wedge b_n$ with atoms on both sides; a *trace* w is a finite list of clauses; its *theory* $\Gamma(w)$ is the set of clauses occurring in it. A *query* is a pair $(H \Rightarrow g)$ of a list of hypothesis atoms and a goal atom. Semantic entailment $\Gamma \models (H \Rightarrow g)$ holds when every model of Γ that makes H true makes g true. *Logical identity* is $u \equiv_L w$ iff $\Gamma(u) \models q \leftrightarrow \Gamma(w) \models q$ for every query q ; it is a congruence for concatenation, and permuting, repeating, or appending a derivable clause to a trace preserves it.

A *recognizer* is a map $\rho(w, q)$ from traces and queries to an observation set. Its *behavioural identity* is $u \approx_\rho w$ iff $\rho(xuz, q) = \rho(xwz, q)$ for all contexts x, z and queries q ; it is a congruence, and the quotient B_ρ is a monoid. The recognizer *respects logic* when $u \equiv_L w$ implies $u \approx_\rho w$, equivalently (the Recognition

Factorization Theorem) when there is a unique monoid morphism $L \rightarrow B_\rho$ commuting with the two quotient maps. The experiments of [2] read language models through Boolean order relations on the answer logits (decision: YES-logit above NO-logit; preference: one query's margin above another's) so that $\approx_\rho$ restricted to a finite test family is an exact identity of Boolean profiles, and found the refinement to fail for every recognizer tested.

The graded closure

Definitions and laws

Let S range over sets of atoms, represented as predicates. One parallel round of forward chaining under Γ is

$$\text{step}_\Gamma(S) = S \cup \{b: (A \rightarrow B) \in \Gamma, A \subseteq S, b \in B\},$$

and the k -round operator is defined by

$$T_0 S = S, \quad T_{k+1} S = \text{step}_\Gamma(T_k S).$$

All statements below are proved in `RecognitionPaths/Graded.lean`; Lean names are given in parentheses.

Proposition 1 (graded-monad laws). For all j, k and S : (i) $T_j(T_k S) = T_{j+k} S$ (`rounds_add`); (ii) $S \subseteq T_k S \subseteq T_{k+d} S$ (`rounds_extensive`, `rounds_le_add`); (iii) T_k is monotone in S and in Γ (`rounds_mono`, `rounds_mono_theory`).

Proof. (i) by induction on j , using $T_{j+1} = \text{step} \circ T_j$ and $(j+1) + k = (j+k) + 1$; (ii) and (iii) by induction on k , since `step` is extensive and monotone in both arguments. ◻

Thus (T_k) is an $\mathbb{N}$ -graded monad on the poset of atom sets, with unit the identity at grade 0 and multiplication the identity $T_j T_k = T_{j+k}$; it is *not* a monad at any single positive grade, since $T_k T_k = T_{2k} \neq T_k$ in general.

Theorem 2 (the limit is the closure). Let $T_\infty S = \bigcup_k T_k S$. (i) *Soundness*: every atom of $T_k S$ holds in every model of Γ that contains S (`rounds_sound`). (ii) *Completeness*: $T_\infty S$ is a model of Γ (`limit_models`); hence $\Gamma \models (H \Rightarrow g) \Leftrightarrow \exists k, g \in T_k H$ (`entails_iff_exists_rounds`).

Proof. (i) by induction on k : a clause whose body holds in the model has its head in the model. (ii) For a clause $A \rightarrow B$ of Γ with $A \subseteq T_\infty S$, each atom of the finite list A lies in some $T_{k_a} S$; by nestedness all lie in $T_K S$ for $K = \sum k_a$, so $B \subseteq T_{K+1} S$. Completeness follows by instantiating the definition of $\models$ at the model $T_\infty H$, which contains H at grade 0. ◻

Graded identity

Write $\Gamma \vdash_k (H \Rightarrow g)$ for $g \in T_k H$ (`EntailsK`). *Budget- k identity* of traces is

$$u \approx_k w \Leftrightarrow \forall q, \Gamma(u) \vdash_k q \leftrightarrow \Gamma(w) \vdash_k q \quad (\text{GradedEquiv}).$$

Proposition 3. For every k : (i) $\approx_k$ is an equivalence relation; (ii) permuting or repeating a trace preserves it (in Lean, `gradedEquiv_of_perm` and `gradedEquiv_dup`); (iii) if $u \approx_k w$ for every k then $u \equiv_L w$ (`logicalEquiv_of_gradedEquiv`), and conversely $u \equiv_L w$ iff for every query the sets of grades at which u and w answer it are both empty or both nonempty (`logicalEquiv_iff_limit`).

Proof. (ii) T_k depends on the trace only through its theory, and $\Gamma(u) = \Gamma(w)$ for permutations and repetitions. (iii) from Theorem 2(ii). ◻

The converse of (iii) fails at every fixed grade, and the failure is exactly characterised:

Theorem 4 (separation of an extension). For a trace w , a clause c and a budget k ,

$$w \approx_k w + [c] \Leftrightarrow \forall q: \Gamma(w + [c]) \vdash_k q \Rightarrow \Gamma(w) \vdash_k q$$

(gradedEquiv_append_iff). Equivalently, w and $w + [c]$ are separated at budget k iff some query is answered within k rounds after the extension and not before it (not_gradedEquiv_append_iff).

Proof. Adding a clause can only add derivations within a fixed budget (monotonicity in Γ), so one implication of $\approx_k$ always holds and the other is the stated condition. ◻

Corollary 5 (transience). If c is derivable from w , then for every query q there is a grade K with $\Gamma(w + [c]) \vdash_{K+d} q \Leftrightarrow \Gamma(w) \vdash_{K+d} q$ for all d (gradedEquiv_append_derivable_eventually).

Proof. If $\Gamma(w) \models q$, take K from completeness; otherwise both sides are false at every grade by soundness and $w + [c] \equiv_L w$. ◻

Theorem 4 says what the budgeted recognizer distinguishes: a redundant extension is visible exactly when it *shortens a derivation across the budget*. Corollary 5 says the visibility disappears once the budget covers the depth. Read as a prediction about an arbitrary recognizer with a budget knob, the theorem asserts a specific interaction between the kind of extension and the budget, and that is what is tested in Section 6.

The closure monad and lax graded readers

The limit of the graded family is, on theories rather than atom sets, the clause closure $\text{Cl } \Gamma = \{c: \Gamma \models c\}$ (Closure.lean). It is extensive, monotone and idempotent (cl_extensive, cl_mono, cl_idem), hence a monad on the poset of theories; its algebras are the closed theories, and

Proposition 6 (logicalEquiv_iff_cl_eq). $u \equiv_L w$ iff $\text{Cl } \Gamma(u) = \text{Cl } \Gamma(w)$.

So the logical meaning space L of [1] is the set of algebras of this monad, and (T_k) is its filtration by rounds (mem_cl_iff_exists_rounds).

A trained reader need not compute T_k . Define a *lax graded reader of slack s* (LaxGraded Γ s) as a family of monotone operators R_k on atom sets with $T_{k-s}S \subseteq R_kS \subseteq T_kS$. Then (Lean names in parentheses): slack 0 forces $R_k = T_k$ (eq_rounds_of_slack_zero); $R_0 = \text{id}$ for every slack (zero_eq_id); and composition inherits the sandwich with the slack added,

$$T_{j+k-2s}S \subseteq R_j(R_kS) \subseteq T_{j+k}S \quad (\text{comp_sandwich}).$$

The proofs are two applications of monotonicity and the composition law of T . The point of the definition is that its hypotheses (monotonicity, slack) and its conclusion (the composite's slack) are all measurable on a learned reader, so the theorem becomes a prediction about the reader's own multiplication.

The budgeted recognizer and the composition law

Let $\rho_k(w, q) = [\Gamma(w) \vdash_k q]$ (roundsRecognizer). It is a theory-factoring recognizer in the sense of [1], so it is invariant under permutation and repetition of any block in any context, and its behavioural identity is $\approx_k$ in every context: $u \approx_{\rho_k} w \Leftrightarrow \forall x, z, xuz \approx_k xwz$ (roundsRecognizer_contextEquiv_iff).

Proposition 7 (composition law on inputs). For all j, k , hypotheses H and goal g :

$g \in T_k(T_jH) \Leftrightarrow g \in T_{j+k}H$ (entailsK_presaturate). On the recognizer,

$$\rho_k(w, T_jH \Rightarrow g) = \rho_{j+k}(w, H \Rightarrow g):$$

pre-saturating the hypotheses by j rounds and reading with budget k is reading with budget $j + k$. A hint buys exactly its depth.

Proposition 7 is a restatement of Proposition 1(i), but as a statement about inputs it is testable on any recognizer that accepts a hypothesis set and has a budget knob, and it is not implied by the locality bound that makes Theorem 4’s pattern available to a message-passing architecture: a model may derive correctly from single atoms at every budget and still fail to use a set of hypotheses as the algebra requires. It is therefore the second, independent prediction.

Predictions and preregistration

Both predictions were committed, with the rules below, before any learned recognizer was run on the corresponding table; the symbolic reasoner and the oracle were run first as deterministic control validation.

P1 (identification). Take a base theory D with a target query $(a \Rightarrow g)$ of depth $d \geq 3$. Let F append a derivable clause that shortens the target’s depth by the least possible amount, and C append a derivable clause that preserves it; $D \equiv_L F \equiv_L C$ by certified closure equality. For a recognizer with budget k let $\text{dis}_X(k)$ be the fraction of cases whose target decision differs between D and X , and $\Delta(k) = \text{dis}_F(k) - \text{dis}_C(k)$. Theorem 4 predicts $\Delta(k) > 0$ for k below the base depth and $\Delta(k) = 0$ once k covers it. The primary comparison is the interaction $I = \Delta(2) - \Delta(4)$ for a learned reasoner trained at four rounds. *Criterion:* $I > 0$ with a paired case-bootstrap 95% confidence interval ($B = 5000$) excluding 0, and $\Delta(2) \geq 0.10$. *Controls:* the exact oracle and a constant recognizer (both $\Delta = 0$), the k -round symbolic reasoner (the theorem by construction), Pythia-70M as an empirical non-reader, and a logic-change condition L (one clause deleted so the target becomes underivable) that every reading recognizer must separate.

P3 (composition). On the same weights, with hypothesis sets $H_j = T_j\{a\}$ for $j \in \{0, 1, 2\}$, the agreement rate between $\rho(H_j; k)$ and $\rho(H_0; j + k)$ over cases must have bootstrap lower bound ≥ 0.95 on each of the four pairs $(j, k) \in \{(1, 2), (2, 1), (2, 2), (1, 3)\}$, chosen because the symbolic yes-rate on the target changes between budgets k and $j + k$ there, so that a constant answer cannot pass.

P5 and P6 (a language model with a thinking budget). P5 is P1 read on Gemini 3.1 Flash-Lite with budgets B0 (no thinking) and B1 (thinking budget 1024), decision readout from the generated YES/NO, gated by accuracy ≥ 0.75 and $\text{dis}_L \geq 0.5$ at each budget: $I = \Delta(B0) - \Delta(B1) > 0$ with CI excluding 0 and $\Delta(B0) \geq 0.10$. After P5 failed with an exploratory directional pattern, P6 was preregistered on the fresh RQ2c table: (a) the net accuracy change of F at B0 (fixes minus breaks on the target) is $\geq +0.05$ with CI above 0; (b) that of C is ≤ -0.05 with CI below 0; (c) at B1 $\text{dis}_F, \text{dis}_C \leq 0.02$.

P9 and P10 (the reader’s own composition). Let $R_k\{a\} = \{a\} \cup \{g' \neq a : \rho(\{a\}, g'; k) = \text{YES}\}$ be the learned reader’s own output, and feed it back: $\rho(R_k\{a\}, g; j)$. P9: for all $(j, k) \in \{1, 2, 3\}^2$ the slack-2 sandwich $[g \in T_{j+k-2}\{a\} \Rightarrow \text{YES}] \wedge [\text{YES} \Rightarrow g \in T_{j+k}\{a\}]$ holds with bootstrap lower bound ≥ 0.95 (what comp_sandwich guarantees if the reader is monotone with slack 1). P10: monotonicity, the rate of goals answered YES from $\{a\}$ and NO from $\{a, y\}$, is ≤ 0.05 at each budget. The unit law $R_0 = \text{id}$ is reported, not predicted.

P7 and P8 (dose–response). On the fresh table at B0: P7a, the net effect of three depth-preserving redundant clauses is more negative than that of one (paired CI excluding 0); P7b, a three-way discrimination on $\delta = \text{net}(N_1) - \text{net}(C_1)$, where N_1 adds one irrelevant clause (not derivable, and invisible from the hypothesis): $|\delta| \leq 0.05$ “count”, $\delta > 0.05$ “redundancy”, $\delta < -0.05$ “distractive”, with “count” stated as the expectation. P8: with an intermediate budget B256 (about 140 thinking tokens used), the effects of F and C_1 shrink relative to B0 with CIs excluding 0 and do not reverse sign.

P4 (sandwich). After P3 failed with all violations in one direction, a third table of 200 cases was generated from a new seed and the weakened law was preregistered: $g \in T_{j+k-1}\{a\}$ implies $\rho(H_j; k) = \text{YES}$, and $\rho(H_j; k) = \text{YES}$ implies $g \in T_{j+k}\{a\}$. Criterion: the per-case rate at which both bounds hold has bootstrap lower bound ≥ 0.95 on every pair $j \in \{1, 2\}$, $k \in \{1, 2, 3, 4\}$.

Sample size: 200 logical structures; for proportion differences near 0.15 the paired bootstrap half-width is about 0.05, and for agreement near 0.97 about 0.025. Both preregistrations state the sentence to be reported on failure; the one for P3 is reported verbatim in Section 6.3.

Instruments

Tables

Seven atoms $a, \dots, g$. A base theory is five random Horn clauses (70% single-body single-head, 15% two-body, 15% two-head) with target hypothesis a and goal the derivable atom of maximal depth, redrawn until the depth is at least 3. Candidate extensions are single-atom clauses $x \rightarrow y$ with $y \in T_\infty\{x\}$, not already a sub-clause of the theory. F is the sorted-first candidate among those that reduce the target depth by the least amount; F_1 is the direct clause $a \rightarrow g$ (maximal shortening and maximal lexical overlap with the query, a secondary condition); C is the sorted-first depth-preserving candidate with the same overlap with $\{a, g\}$ as F when one exists (181 of 200 cases); L deletes the sorted-first clause whose removal makes the target underivable. Cases without an F , a C , an L , a depth-1 atom or a non-derivable atom are dropped and the next draw used: 200 cases from 264 draws, one seed, taken in order. Realised depths: base 3 in 177 cases and 4 in 23; F reduces to 2 in 176, 3 in 23, 1 in 1. Four queries per case: the target t , a depth-1 atom d_1 , a non-derivable atom n , and the reversed query r . Rows: $200 \times 5 \times 4 = 4000$. The composition table takes the 200 base theories with $H_0 = \{a\}$, $H_1 = T_1\{a\}$ ($|H_1| \in \{2, 3, 4\}$), $H_2 = T_2\{a\}$ ($|H_2| \in \{3, 4, 5\}$), for t and n : 1200 rows. Both tables carry SHA-256 locks.

Recognizers

Symbolic. The exact closure oracle; the k -round reasoner $[g \in T_k H]$ for $k \in \{0, \dots, 6\}$; a constant-NO recognizer.

Iterative learned reasoner. Each atom carries a state vector ($d = 64$), initialised from a learned embedding plus a learned flag on the hypothesis atoms. In each round every clause computes a message from the sum and the elementwise minimum of its body states and the current head state (a two-layer MLP), messages are aggregated per head atom by an elementwise maximum (so the model is invariant to clause permutation and repetition by construction), and each atom’s state is updated by a GRU cell. After k rounds the goal state and the maximum over hypothesis states are read out by an MLP into two logits. The number of rounds is the budget and can be set at evaluation. 75,202 parameters. Trained 6000 steps (batch 128, AdamW, one-cycle schedule) on random theories of two to six clauses over seven atoms with balanced labels and random atom relabelling, with every evaluation theory (D, F, F_1, C, L of every case) excluded from training up to relabelling. The primary model is trained with a budget of four rounds (in-distribution validation accuracy 0.996); a secondary model with two rounds (0.960). Both were trained with single-atom hypotheses; the composition test feeds them hypothesis sets through the same flag, out of the training format (for a single atom the modified readout is numerically identical to the trained one).

Set recognizer. A seven-atom version of the max-pooled clause encoder of [2] (168,386 parameters, validation accuracy 0.951); no budget knob.

Open-weight language models. Qwen2.5-0.5B-Instruct and Qwen2.5-1.5B-Instruct, raw prompt (no chat template), bullet rendering of the clauses with per-case atom names, one forward pass on CPU in float32, observation the pair of logits of the tokens YES and NO at the answer position; Pythia-70M (step 143000) as the non-reading control.

A language model with a thinking budget. Gemini 3.1 Flash-Lite through its API, temperature 0, the same prompt, at budget B0 (no thinking configuration; the model emits the answer directly) and B1 (thinking budget 1024 tokens; about 400 used on average). Logprobs are not exposed for this model, so the observation is the generated text and the decision its first token, YES or NO (unparseable, 3 of 11,200 requests, counted as NO). Rows: the RQ2 rows for queries t and n and the pre-saturated rows for $j \in \{1, 2\}$, on the first table (RQ2d) and on the fresh table (RQ2e), 5600 requests each. The dose-response table (RQ2f) adds, on the fresh cases, C_2 and C_3 (C_1 plus the next one or two sorted-first depth-preserving derivable clauses; 155 and 106 cases have them), N_1 (198 cases), and a budget of 256 tokens for D, F, F_1, C_1, L ; 2259 requests. A pilot showed the model exposes three effective thinking levels, 0, about 140 (budgets 256 and 512) and about 300 tokens (1024 and 2048).

Readout

Decisions are Boolean (positive logit strictly above negative; ties count as NO). The extended observation is the real margin. All statistics are over the 200 cases with paired case bootstrap.

Results

P1 holds, and the learned reasoner is the symbolic one

Table 1 gives the primary model at each budget.

budget k	$\text{acc}(D)$, target	dis_F	dis_{F_1}	dis_C	dis_L	Δ [95% CI]
1	0.00	0.01	1.00	0.00	0.00	0.01 [0.00, 0.01]
2	0.00	0.89	1.00	0.00	0.00	0.89 [0.84, 0.93]
3	0.89	0.11	0.11	0.01	0.89	0.10 [0.06, 0.15]
4	1.00	0.00	0.00	0.00	1.00	0.00 [0.00, 0.00]
6	1.00	0.00	0.00	0.00	1.00	0.00 [0.00, 0.00]

Table 1. The four-round learned reasoner on 200 cases; decision-change rates relative to the base D on the target query. $I = \Delta(2) - \Delta(4) = 0.885$, 95% CI [0.84, 0.93]. **P1 holds.**

The learned reasoner’s decision table coincides with that of the k -round symbolic reasoner at every budget (Table 9 in the appendix lists both). At $k = 2$ the 177 depth-3 bases are NO and their F extensions (depth 2) YES; the 23 depth-4 bases are NO on both. At $k = 3$ the depth-4 bases separate (0.11) and at $k = 4$ nothing does except the logic change. Every D – F disagreement is F correct and D wrong; the reverse never occurs. The direct clause F_1 separates already at $k = 1$, because it is a depth-1 derivation, and not at $k = 4$, so lexical overlap with the query plays no role for this recognizer. The controls behave as required: oracle and

constant recognizer give $\Delta = 0$; Pythia-70M gives $\Delta = -0.02$ $[-0.04, 0.01]$ and $\text{dis}_L = 0.04$, i.e. it does not read.

A caveat stated in the preregistration: information in this architecture flows one hop per round, so a *correct* model must show this pattern. The empirical content is that training produced this reader rather than a shortcut (no clause-count or lexical heuristic survives at $k = 2$), that the gap closes exactly at the predicted budget, and that the pattern is what Theorem 4 says and nothing more.

Secondary constructed recognizers

The two-round model. Trained under a budget too small for its data, it shows the same interaction, $I = 0.665$ $[0.60, 0.74]$, and closes the gap at $k = 4$ although it never ran four rounds in training (its rounds extrapolate; at $k = 6$ accuracy degrades to 0.90 and the logic change becomes partly invisible, $\text{dis}_L = 0.70$). It differs from the primary model in one informative way: at its own budget it says YES on 16% of the depth-3 bases it cannot derive, and its decision moves on 10% of the depth-preserving extensions C . A model trained under an insufficient budget acquires a shortcut component that a merely redundant clause can trigger; depth accounts for 0.67 of its 0.77 F effect.

The set recognizer. With no budget it identifies almost every extension at the decision level ($\text{dis}_F = 0.01$, $\text{dis}_{F_1} = 0.00$, $\text{dis}_C = 0.00$) and reads the logic change ($\text{dis}_L = 0.94$). Its margins nevertheless order the conditions $F_1 > F > C > 0 > L$ (mean shifts on the target $+1.91$, $+0.65$, $+0.23$, -9.3 logits, with $F - C = 0.42$ $[0.29, 0.54]$): a recognizer without rounds still orders redundant extensions by depth shortening in its extended observation.

P3 fails for the primary model

pair (j, k)	symbolic	four-round model (primary)	two-round model
(1, 2)	1.000	0.897 [0.870, 0.925]	0.990 [0.980, 0.998]
(2, 1)	1.000	0.830 [0.797, 0.863]	0.975 [0.960, 0.990]
(2, 2)	1.000	0.995 [0.988, 1.000]	0.998 [0.993, 1.000]
(1, 3)	1.000	0.983 [0.970, 0.995]	1.000 [1.000, 1.000]

Table 2. Agreement between $\rho(H_j; k)$ and $\rho(H_0; j + k)$ over 200 cases and two queries, with paired bootstrap 95% CI.

Two of the four primary pairs fall below the criterion, so **P3 fails** for the primary model. As preregistered, we report: *on 200 fresh Horn cases the learned reasoner did not satisfy the composition law $T_k \circ T_j = T_{j+k}$ out of its training format (agreement 0.897 [0.870, 0.925] at $j = 1, k = 2$ and 0.830 [0.797, 0.863] at $j = 2, k = 1$); the graded-monad description fits its budget behaviour but not its hypothesis-set behaviour, and the law is a property of the symbolic reasoner that the learned one does not inherit.*

The failure has one direction. In all 41 disagreements at (1, 2) and all 68 at (2, 1) the model answers NO from the pre-saturated hypotheses at the tight budget where the law says YES; it never answers YES where the law says NO. It reads multi-atom hypothesis sets correctly when given slack (accuracy 1.000 at $k = 4$ with H_1 and with H_2); agreement falls with the size of the hint (at (2, 1): $|H_2| = 3$: 0.73, 4: 0.63, 5: 0.55) and is near-perfect on the depth-4 cases (0.957), where a spare round exists. The learned operator therefore satisfies the lax inequality $T_k \circ T_j \subseteq T_{j+k}$, with strict inclusion on 20–35% of cases at the tight budgets: it loses part of a round when a derivation starts from a set rather than an atom. The two-round model, which

had to propagate from whatever it was given within two rounds, passes all four pairs; this was not preregistered for it and is reported as an observation.

P4 holds: the one-round sandwich law on a fresh seed

pair (j, k)	sandwich, 4-round [95% CI]	exact law, 4-round	sandwich, 2-round	exact law, 2-round
(1, 1)	1.000 [1.000, 1.000]	1.000	0.993	0.892
(1, 2)	1.000 [1.000, 1.000]	0.895	0.993	0.993
(1, 3)	1.000 [1.000, 1.000]	0.983	0.998	0.998
(1, 4)	1.000 [1.000, 1.000]	1.000	0.998	0.998
(2, 1)	1.000 [1.000, 1.000]	0.818	1.000	0.990
(2, 2)	1.000 [1.000, 1.000]	0.990	0.998	0.998
(2, 3)	1.000 [1.000, 1.000]	0.998	1.000	1.000
(2, 4)	1.000 [1.000, 1.000]	1.000	0.998	0.998

Table 3. The sandwich law $T_{j+k-1} \subseteq \rho(H_j; \cdot; k) \subseteq T_{j+k}$ and the exact composition law on 200 cases from a second seed (never seen by any analysis), two queries per case.

On the fresh table the four-round model satisfies the sandwich law on every case of every pair, while the exact law fails again at (1, 2) and (2, 1) at almost the same rates as on the first table (0.895 vs 0.897, 0.818 vs 0.830). **P4 holds.** The failure of P3 is thereby replaced by a law: the learned reasoner never derives more than the algebra allows and never less than the algebra allows with one round of slack. The two-round model satisfies the sandwich law at ≥ 0.993 (all bootstrap lower bounds ≥ 0.983) and the exact law at ≥ 0.990 on the primary pairs.

P9 and P10 hold; the unit law fails

(j, k)	slack-2 sandwich	slack-1 sandwich	exact = T_{j+k}
(1, 1), (2, 2), (2, 3), (3, 3)	1.000	1.000	1.000
(1, 2)	1.000	1.000	0.818
(1, 3)	1.000	0.890 [0.860, 0.917]	0.870
(2, 1)	1.000	1.000	0.895
(3, 1), (3, 2)	1.000	1.000	0.983, 0.998

Table 4. The 4-round reader fed its own output, 200 fresh cases, mean over the target and the non-derivable atom.

The composite lies within two rounds of slack on every case of every pair (**P9 holds**), and within one round on eight of nine pairs; only one round applied to a three-round output loses a second round, on 11% of cases. Monotonicity violations are 0.000 at every budget (**P10 holds**), so the theorem’s hypothesis is met and its

conclusion is not a coincidence. The unit law fails: at zero rounds the reader agrees with the identity on 0.571 of atoms, a constant answer (it was never trained at zero rounds). The 2-round model’s self-composition agrees with T_{j+k} on ≥ 0.990 of cases on every pair and is equally unitless. What training keeps of the graded monad is therefore a lax, unitless, monotone graded family inside it; what it does not keep is the unit and the exactness of the multiplication.

Language models

On this rendering (seven atoms, five or six clauses) Qwen2.5-0.5B answers YES on every query of every condition (minimum margin 1.23) and Qwen2.5-1.5B answers NO on every query, so the decision-level test cannot be evaluated for either; this is the “not reading” gate of the audit protocol in [2] and is reported as not evaluable, not as a refutation. The margins carry partial reading (comparative accuracy $[m_t > m_n]$ on the base: 0.79 and 0.89), so the extended observation can be examined, as preregistered, in an exploratory analysis.

recognizer	$F - C$	$F_1 - F$	L
Pythia-70M (non-reader)	0.010 [0.005, 0.014]	0.004 [−0.001, 0.009]	−0.080
Qwen2.5-0.5B	0.033 [0.026, 0.040]	0.098 [0.091, 0.104]	−0.406
Qwen2.5-1.5B	0.048 [0.030, 0.067]	0.285 [0.268, 0.302]	−0.207
set recognizer	0.417 [0.289, 0.540]	1.264 [1.124, 1.407]	−9.32

Table 5. Mean signed margin shift on the target query relative to the base, in logits, over 200 cases; paired bootstrap 95% CI. Exploratory.

For both language models the shortening clause moves the margin more than the depth-preserving clause, with intervals excluding 0, but by 0.03–0.05 logits: three to five times the non-reader’s surface residual (which exists because F mentions the hypothesis atom in 198 of 200 cases and C in 131; on the 181 overlap-matched cases $F - C$ is 0.025 and 0.029) and an order of magnitude below the direct clause’s effect ($F_1 - F$: 0.10 and 0.29). The variable that predicts a redundant clause’s visibility to these models is lexical overlap with the query; the depth direction is a small residual.

A language model with a thinking budget

	B0, first table	B0, fresh table (preregistered)	B1, both tables
acc(D), target	0.80	0.78	1.00
dis $_F$ / dis $_C$ / dis $_L$	.155 / .140 / .80	.155 / .140 / .78	0 / 0 / 1
unsigned Δ [CI]	0.015 [−0.055, 0.080]	0.015 [−0.050, 0.080]	0
net F (fixes – breaks) [CI]	+0.135 [0.085, 0.190]	+0.155 [0.110, 0.210]	0
net C [CI]	−0.090 [−0.140, −0.040]	−0.070 [−0.120, −0.020]	0
net F_1	+0.200	+0.220	0
signed $F - C$ [CI]	+0.225 [0.165, 0.285]	+0.225 [0.170, 0.285]	0

Table 6. Gemini 3.1 Flash-Lite on the target query, 200 cases per table.

Both budgets pass the reading gate. At B1 the model is the oracle on every row of both tables: accuracy 1.000 on all conditions and queries, $\text{dis}_F = \text{dis}_{F_1} = \text{dis}_C = 0$, $\text{dis}_L = 1$. It is the first recognizer in this

programme, other than the constructed ones at sufficient budget, with an exactly logical decision profile.

At B0 the preregistered P5 **fails**: $\Delta(B0) = 0.015$ with CI covering 0. The failure is instructive. F and C each change about 15% of the decisions, but in opposite directions: the shortening clause turns wrong answers right (fixes 0.145, breaks 0.010) and the depth-preserving clause turns right answers wrong (fixes 0.025, breaks 0.115). The unsigned statistic, designed from the constructed reasoners for which C changes nothing, cannot see this. The signed pattern was therefore preregistered as P6 on the fresh table and **holds** on all three clauses, with the signed difference $F - C$ identical on the two tables (0.225). Secondary observations recur across the tables: the direct clause helps most (+0.20, +0.22); pre-saturation is not monotone at B0 (a one-round hint of two atoms lowers accuracy from 0.80 to 0.71, a two-round hint raises it to 0.93–0.95); at B1 the direct clause saves thinking tokens (−78, −34 relative to D) and the depth-preserving clause costs some (+31, +39), while the minimal shortening saves none.

Dose–response: what the cost is a cost of, and how the budget acts

condition at B0	n	net effect [95% CI]
F (minimal shortening)	200	+0.155 [0.110, 0.210]
F_1 (direct clause)	200	+0.220 [0.165, 0.280]
C_1 (one depth-preserving redundant clause)	200	−0.070 [−0.120, −0.020]
C_2 (two)	155	−0.206 [−0.284, −0.129]
C_3 (three)	106	−0.283 [−0.377, −0.189]
N_1 (one irrelevant non-derivable clause)	198	−0.217 [−0.278, −0.157]

Table 7. Net effect (fixes minus breaks on the target) of each condition relative to D for Gemini 3.1 Flash-Lite without thinking, fresh table.

budget (tokens used)	$\text{acc}(D)$	net F	net F_1	net C_1	dis_L
0	0.78	+0.155	+0.220	−0.070	0.78
256 (≈ 140)	0.80	+0.140	+0.200	−0.070	0.80
1024 (≈ 300)	1.00	0	0	0	1.00

Table 8. The budget response on the same cases.

P7a holds: on the 106 cases with all three doses, $C_1 - 0.057$, $C_2 - 0.217$, $C_3 - 0.283$; $C_3 - C_1 = -0.226$ [−0.311, −0.151]. **P7b falls in the distraction cell**, not the expected count cell: $\delta = -0.146$ [−0.212, −0.081]. An irrelevant clause that the model could not have derived costs about three times as much as a redundant one it could have, and about as much as two redundant ones, although neither touches the target’s derivation. The cost of an added clause is therefore not a cost of length; it depends on the clause’s logical relation to the theory, derivable being cheap and underivable expensive. **P8 fails**: at about 140 thinking tokens nothing changes ($F + 0.140$, $C_1 - 0.070$, accuracy 0.80), and at about 300 the model is the oracle. As preregistered, the budget is not a grade for this model; it is a threshold. The B0 rerun agreed with the RQ2e decisions on every row.

Discussion

What the graded object explains. The companion paper found that every recognizer separates a trace from its redundant extension and could not say why. Theorem 4 gives the reason for one class of recognizers: separation happens exactly when the extension moves a derivation across the budget, and it is transient in the budget. The prediction held with the full shape, including the zero on depth-preserving extensions at every budget, the zero on all extensions at sufficient budget, and the one-sided direction of every disagreement. For the learned reasoner, then, the answer to “which equal-meaning inputs does it distinguish?” is exact and is given by the grade.

Architecture versus training. Two equations of the same object were tested on the same weights. The identification equation is made available by the architecture (one hop per round) and was realised by training. The composition equation is not implied by any locality bound, and the trained model violates it in a specific and now lawful way: it is exactly graded in what it separates and laxly graded in how it composes, with $T_{j+k-1} \subseteq \rho(H_j; k) \subseteq T_{j+k}$ on every fresh case. A model trained under a smaller budget satisfies the equation itself. The slack is therefore a training quantity with a definite size, one round, and not a failure of the description. The graded monad is therefore the right description of the *quotient* the recognizer computes; whether the recognizer also carries the monad’s *multiplication* depends on how it was trained. In the programme’s terms, the algebraic theory has been reached for one operation and refuted for one equation, by a test of the equation and not by construction.

The language-model variable. For the two open-weight models the graded account is not what predicts sensitivity; lexical overlap with the query is. For the model with a thinking budget the picture is different and sharper. With budget it is the ideal recognizer, so there is nothing for the graded account to separate. Without budget the shortening clause helps, as the graded account predicts, and any clause that does not shorten hurts, which it does not predict, in proportion to how many are added and more when the clause is underivable than when it is redundant. The second term is not lexical (the clauses are matched in shape and overlap) and not length (a derivable clause is cheap, an irrelevant one expensive); it is a logical property of the added clause relative to the theory, seen by a recognizer that does not compute the closure. Both terms vanish at once, in a single step of the thinking budget. That the opposite-signed effects cancel in the unsigned statistic, and were found only because a failure was examined and then retested under preregistration, twice, is the methodological point of the paper.

The monad question. The programme asked for the monad only after the operations and equations. For the budgeted reasoner they are now known: an $\mathbb{N}$ -graded family of monotone operators, bounded above by the graded closure and below by its previous grade, composing with slack at most two (usually one), and without unit. The closure monad and its filtration are the objects the reader approximates; the approximation is lax and unitless, and both of those facts were established by preregistered tests rather than assumed. For the language models no monad is named: at zero budget Gemini’s behaviour is not a quotient of traces by a congruence (a derivable and an underivable clause act differently on the same consequence set), and no finite test family has been found closed for any language model. That is the honest state of the question.

What remains untested. The composition law is an equation between budgets in rounds; thinking tokens are a budget but not in rounds, and for this model they act as a threshold, so only qualitative consequences were examined here. Pre-saturation was not monotone at zero budget on either table (a two-atom hint hurts, a larger hint helps), which no reading of the graded structure predicts and which is reported as an open observation. A budget knob calibrated in rounds for a language model, or a language model whose thinking can be read as forward chaining, would allow the sandwich law to be tested there as it was for the learned reasoner.

Limitations

Two open-weight language models, both under two billion parameters, one rendering, no chat template, one forward pass; their decision-level tests are not evaluable and the margin-level analysis is exploratory. One closed model with a thinking budget, three effective budget levels, a text-only decision readout, and no access to its reasoning; the signed effects are confirmed on one fresh table and the dose–response results on the same fresh table, and no larger claim is made. The constructed models are small and the iterative reasoner’s architecture makes the identification pattern available to a correct model, as stated in advance; the composition result for the two-round model was not preregistered. Seven atoms and five-clause bases bound the depths to 3 and 4; the budgets compared are therefore 2 against 4, and monotone behaviour across all budgets was not assumed or claimed. The Lean development is elementary and is offered to fix definitions, not as new mathematics. No claim is made that any recognizer, learned or linguistic, “has” a monad; the paper’s claims are that a named algebraic object predicts one behaviour of a budgeted recognizer exactly, fails to predict another, and does not predict the language models examined.

Artifacts and reproducibility

Theory: RecognitionPaths/Graded.lean in the repository recognition-paths (Lean 4 v4.24.0, no Mathlib; concept DOI 10.5281/zenodo.22495808). Tables, generators, runners, analysis and preregistrations: rq2/ and docs/ in proof-path-invariance (concept DOI 10.5281/zenodo.22495706). The frozen tables have the SHA-256 locks

```
rq2_prompts.jsonl
39a82a17f9aae4027586339c58942a0114439bf0f978cf12724e754479676510
rq2b_presat.jsonl
5e42376dd5e01a7f4f6b04856a74a4596f8378ea6579786acce489928456a910
table_c/rq2_prompts.jsonl
a65f577afce5aecab424e4fa8def107144a4da276c0942001a0f17c0ec51ed
table_c/rq2c_presat.jsonl
346ea642db4d19d6667b0ef83589ba1492748e74c5a49626b0e7fd7ecd8bfb7e
rq2d_llm.jsonl
6f2375f5bfe4c7eb585477840699409fd38d6be5c2757246e309456c053e4abe
table_c/rq2e_llm.jsonl
211e2a7f32aea9a4756ce7ccfe1972d24cf2f9ca50b9bf6019c2af21974f29b2
table_c/rq2f_dose.jsonl
3ddef4d5044ce8de4f261199632fd0ce959c6953a5f4d8e9227be5f4eacb05fc
```

Weights and training logs: jinu0633/recognition-paths-recognizers, folder rq2/ (Hugging Face). The paper repository graded-recognition carries the manuscript, the exact commands, verbatim copies of the preregistrations (RQ2, RQ2b–g) with their outcomes, and docs/MONAD.md, the record of which algebraic laws each recognizer satisfies. All runs are CPU-only.

Appendix: control validation and full budget tables

budget k	symbolic $\text{acc}(D)$	symbolic $\text{dis}_F / \text{dis}_C / \text{dis}_L$	learned $\text{acc}(D)$	learned $\text{dis}_F / \text{dis}_C / \text{dis}_L$
1	0.75	0.01 / 0.00 / 0.00	0.75	0.01 / 0.00 / 0.00
2	0.75	0.89 / 0.00 / 0.00	0.75	0.89 / 0.00 / 0.00
3	0.97	0.12 / 0.00 / 0.89	0.97	0.11 / 0.01 / 0.89
4	1.00	0.00 / 0.00 / 1.00	1.00	0.00 / 0.00 / 1.00
6	1.00	0.00 / 0.00 / 1.00	1.00	0.00 / 0.00 / 1.00

Table 9. The k -round symbolic reasoner and the four-round learned reasoner on the same table; accuracy is over all four queries.

pair (j, k)	yes-rate $(H_j; k)$	yes-rate $(H_0; j + k)$	agreement on t	agreement on n
(1, 1)	0.000	0.000	1.000	1.000
(1, 2)	0.685	0.890	0.795	1.000
(2, 1)	0.550	0.890	0.660	1.000
(2, 2)	0.990	1.000	0.990	1.000
(1, 3)	0.965	1.000	0.965	1.000
(1, 4), (2, 3), (2, 4)	1.000	1.000	1.000	1.000

Table 10. The composition test for the four-round model by query; the symbolic reasoner agrees at 1.000 on every pair and the two-round model at 0.975 or above on every primary pair.